

Micro:bit → Teachable Machine

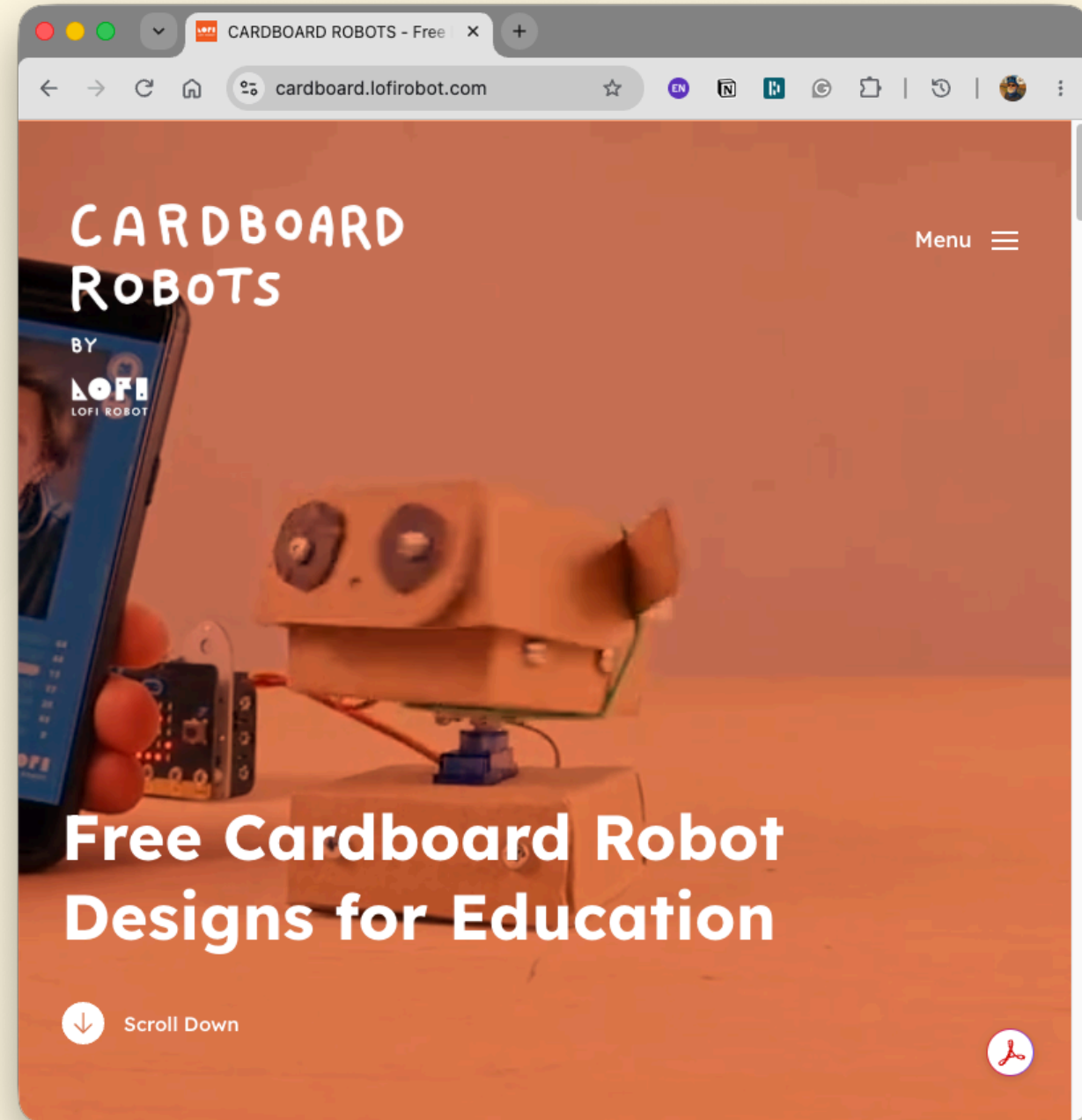
Micro:bit → LLM Claude

Workshop



Inspired from Carboard Robots

- By LOFI Robots - Maciej Wojnicki
- Polish Educator and Fab Lab Pioneer
- <https://cardboard.lofirobot.com/>



LoFi Robotics Instructions

- <https://cardboard.lofirobot.com/teachable-microbit-app-info/>



Setup Micro:Bit 1

To make the board discoverable over bluetooth you have to properly set PROJECT SETTINGS in MakeCode (individual for each project!)

1. go to (GEAR ICON) -> PROJECT SETTINGS
2. Set project settings sliders to NO PAIRING
3. Next click EDIT SETTINGS AS TEXT
4. scroll down to section about BLUETOOTH and add this line:
 “pairing_mode”: 0,

BlocksJavaScript



Sign In

Go back

Name

blockout

☒ No Pairing Required: Anyone can connect via Bluetooth.

☐ JustWorks pairing (default): Pairing is automatic once the pairing is initiated.

☐ Passkey pairing: Pairing requires 6 digit key to pair.

☐ Disable Bluetooth Event Service

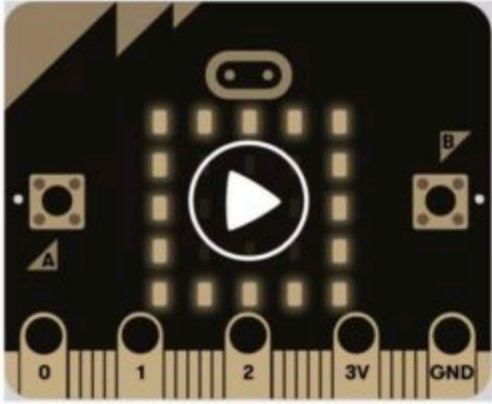
Save

Edit Settings As text

Microsoft | micro:bit

BlocksJavaScript

HomeShareHelpSettingsSign In



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Show data Device

Explorer

```
14  "targetVersions": {
15      "branch": "v1.1.61",
16      "tag": "v1.1.61",
17      "commits": "https://github.com/Microsoft/pxt-microbit/commits/60286e3cfd2ea167a1de5d55f06",
18      "target": "1.1.61",
19      "pxt": "4.1.12"
20  },
21  "preferredEditor": "blocksprj",
22  "yotta": {
23      "config": {
24          "microbit-dal": {
25              "bluetooth": {
26                  "open": 1,
27                  "pairing_mode": 0,
28                  "whitelist": 0
29              }
30          }
31      }
32  }
33  }
34  }
```

Download

Teachable Micro:Bit - bi

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Setup Micro:Bit 2

```
"config": {  
    "microbit-dal": {  
        "bluetooth": {  
            "open": 1,  
            "pairing_mode": 0,  
            "whitelist": 0,  
            "security_level": null  
        }  
    }  
}
```

VERY Important Notes!

- In Teachable Machine model use class names without spaces. If you need multi-word class names, join them with underscores, like this:
`this_class_name`
- Teachable Micro:Bit App gets cached in browser memory and to prevent it open the app in new incognito window.

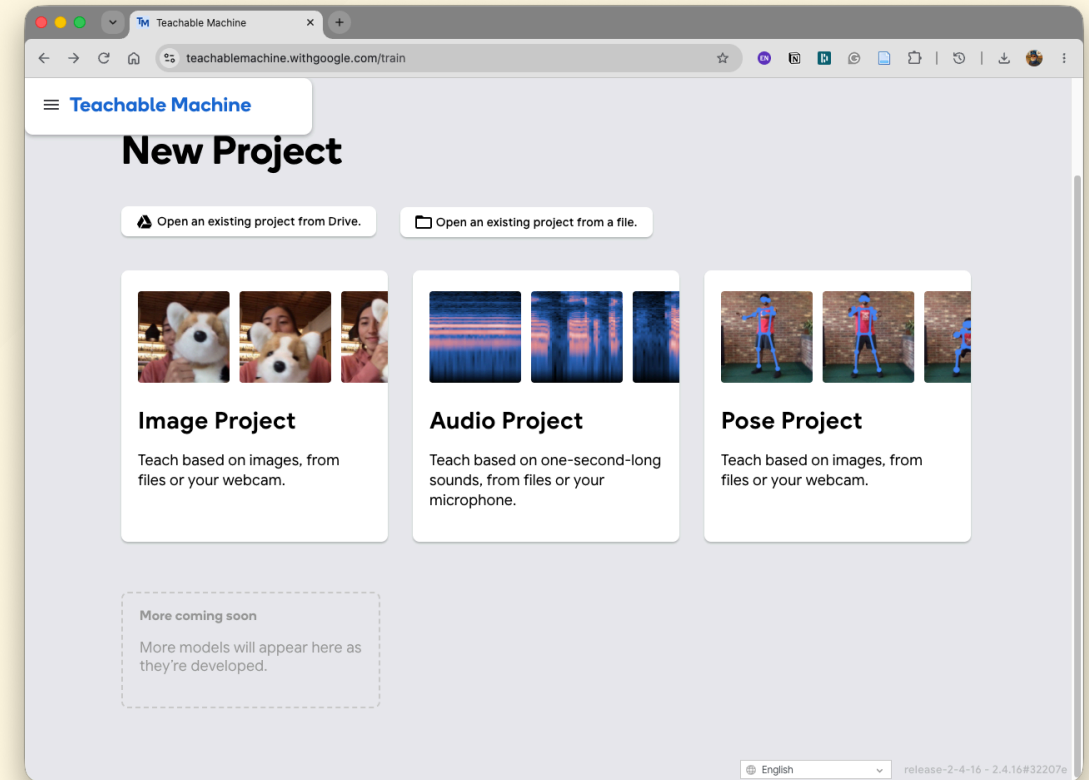
Saving models links

You can pass it as a page address parameter, for example:

- Link to TM model:
<https://teachablemachine.withgoogle.com/models/Jgyr546VH/>
- Take the last part `Jgyr546VH` and add it to the microbit app address like that:
<https://cardboard.lofirobot.com/teachable-microbit/?model=Jgyr546VH>
- This way the model will load automatically after opening the app and you can access the model simply by storing this link.

Teachable Machine

<https://teachablemachine.withgoogle.com/train>



TM Image Model - Teachable Mac

teachablemachine.withgoogle.com/train/image/1Di8wQxqOcRzt_8HGnhUtMy4hU-GOx0kC

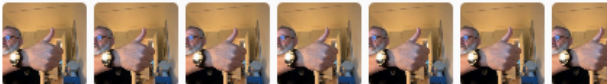
Teachable Machine

OK

27 Image Samples

Webcam

Upload

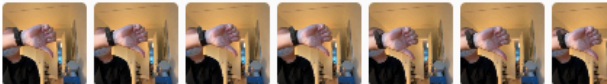


NOT

13 Image Samples

Webcam

Upload



+

 Add a class

Training

Train Model

Advanced

Preview

Export Model

You must train a model on the left before you can preview it here.

English release-2-4-16 - 2.4.16#32207e

Teachable Machine

27 Image Samples

Webcam Upload

13 Image Samples

Webcam Upload

Export your model to use it in projects.

Tensorflow.js TensorFlow TensorFlow Lite

Export your model:

☒ Upload (shareable link) ☐ Download

Your sharable link:

https://teachablemachine.withgoogle.com/models/iM_HPnNx3/ Copy

When you upload your model, Teachable Machine hosts it at this link. (FAQ: [Who can use my model?](#))

✓ Your cloud model is up to date.

Code snippets to use your model:

JavaScript p5.js [Contribute on Github](#)

Learn more about how to use the code snippet on [github](#).

```
<div>Teachable Machine Image Model</div>
<button type="button" onclick="init()">Start</button>
<div id="webcam-container"></div>
<div id="label-container"></div>
<script src="https://cdn.jsdelivr.net/npm/@tensorflow/tfjs@latest/dist/tf.min.js"></script>
<script src="https://cdn.jsdelivr.net/npm/@teachablemachine/image@latest/dist/teachablemachine-
image.min.js"></script>
<script type="text/javascript">
  // More API functions here
```

Copy

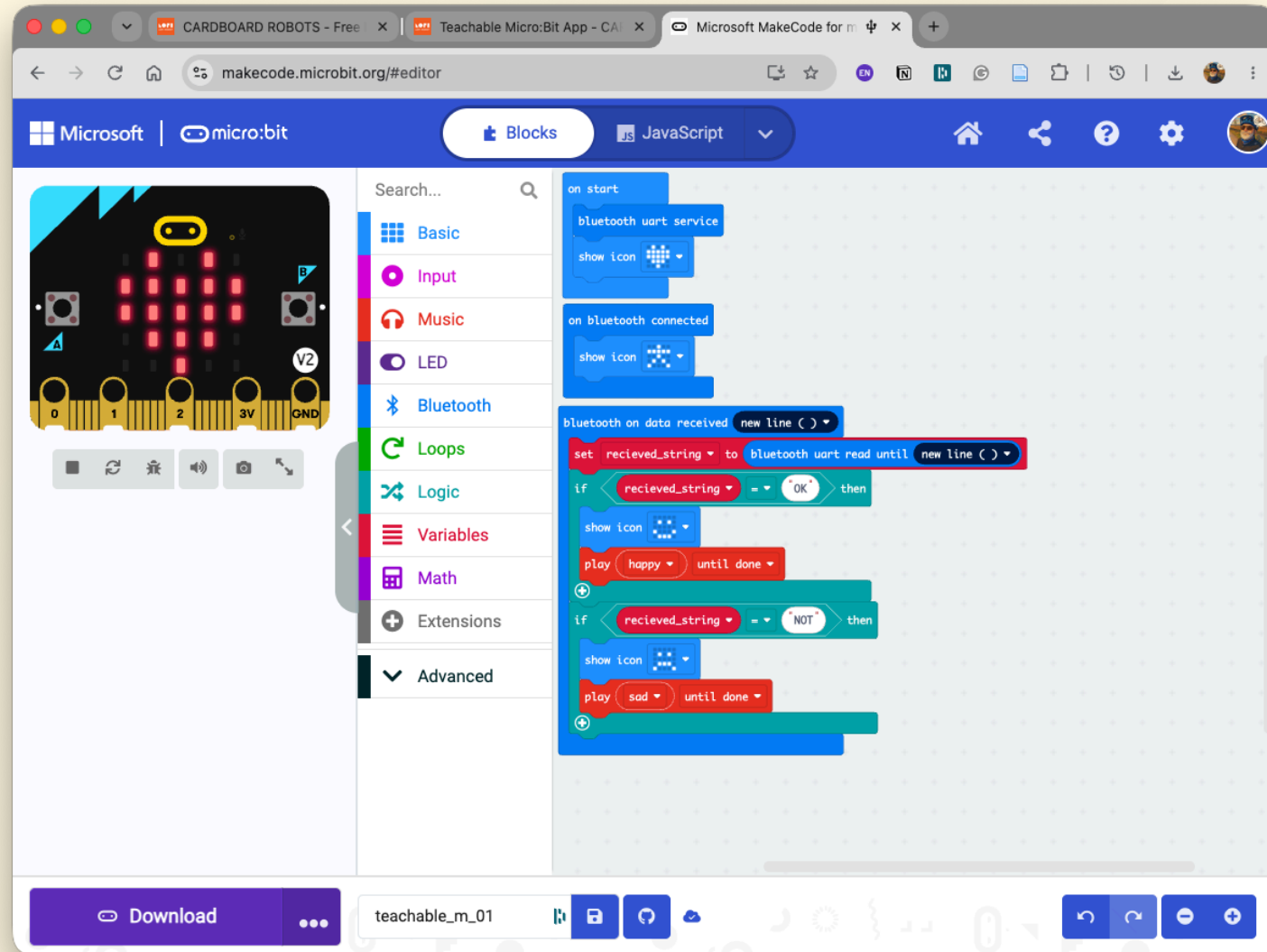
Export Model

Model is working

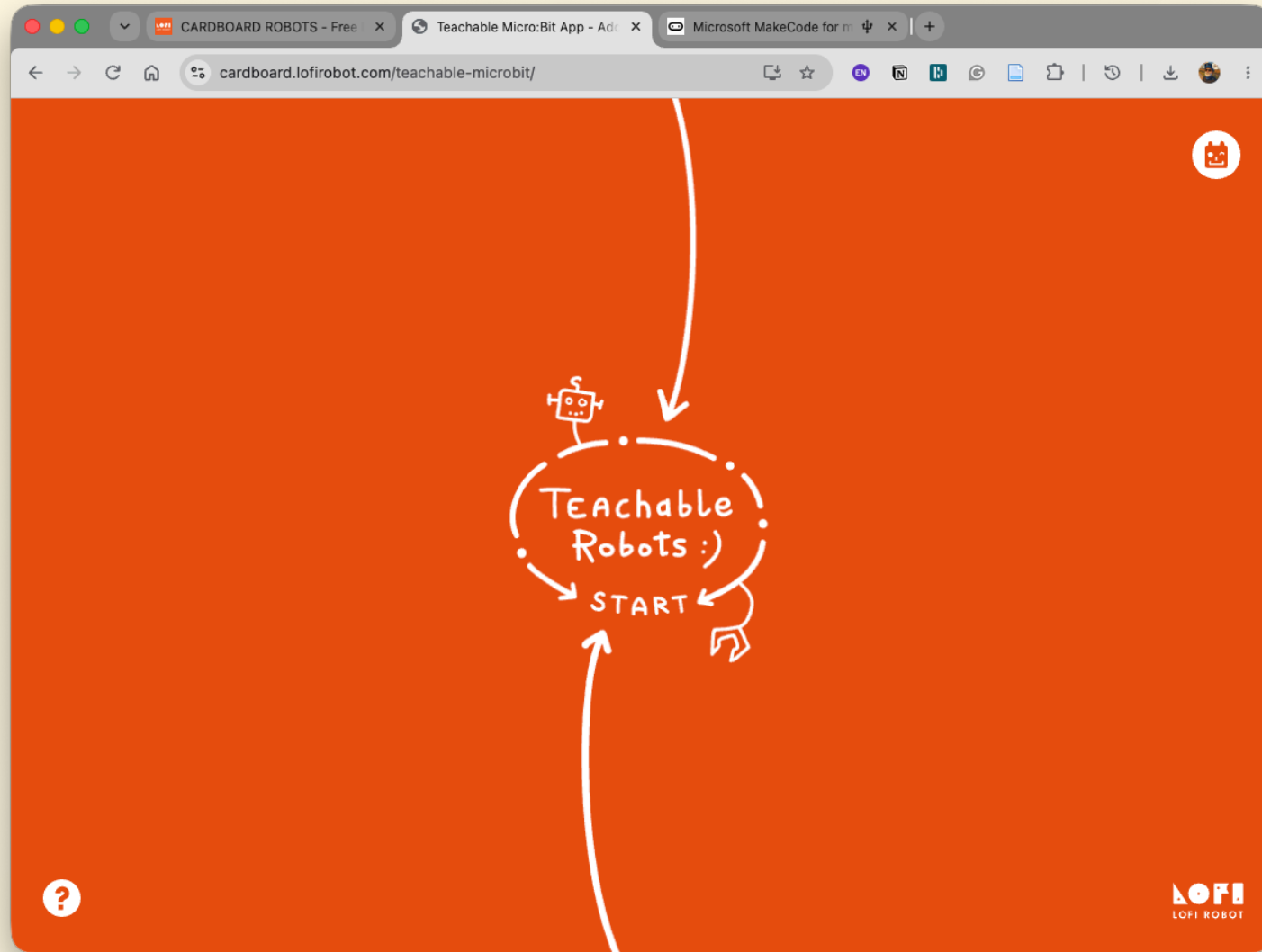
59%

41%

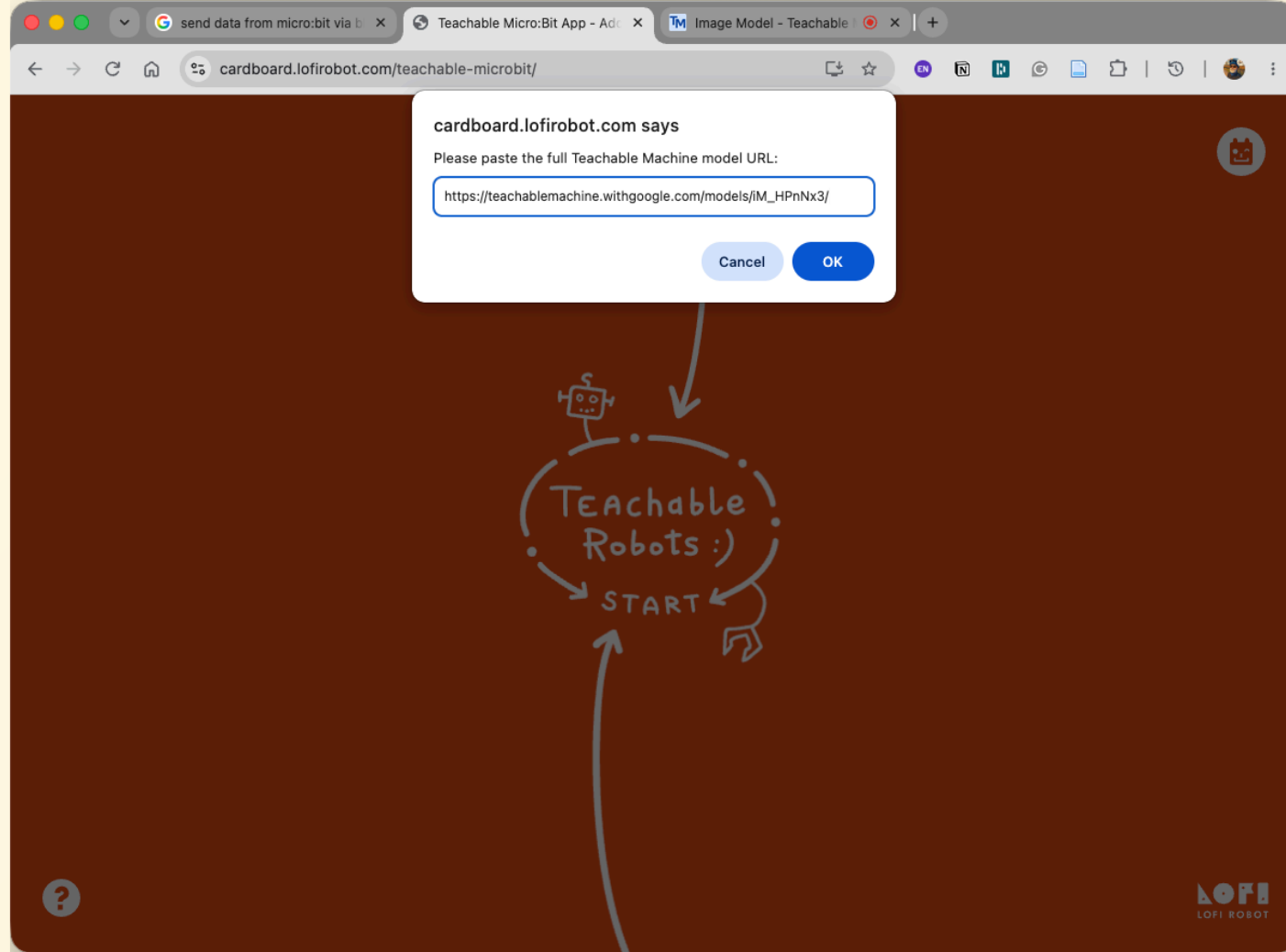
Make Code Blocks



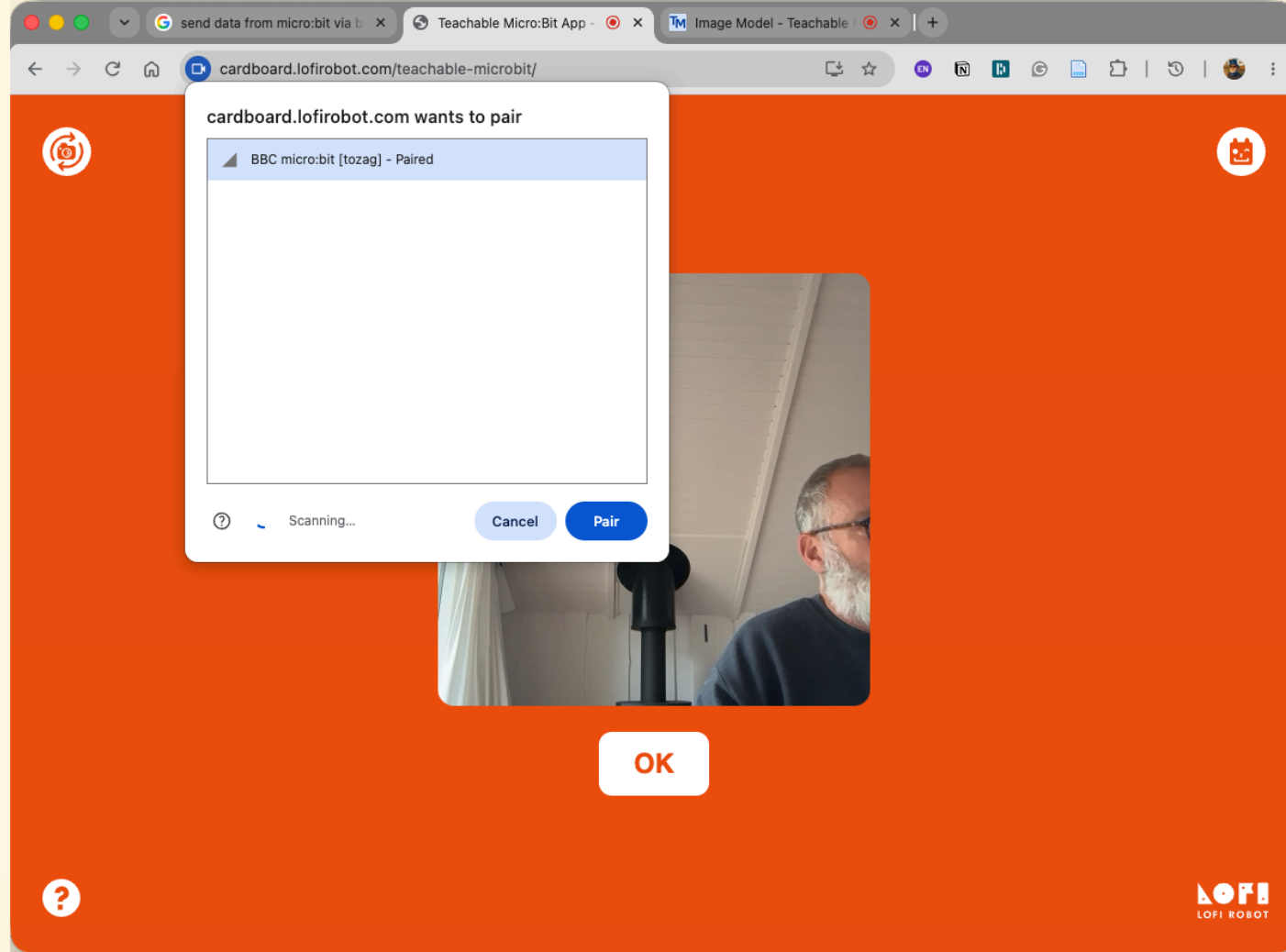
Lofi APP



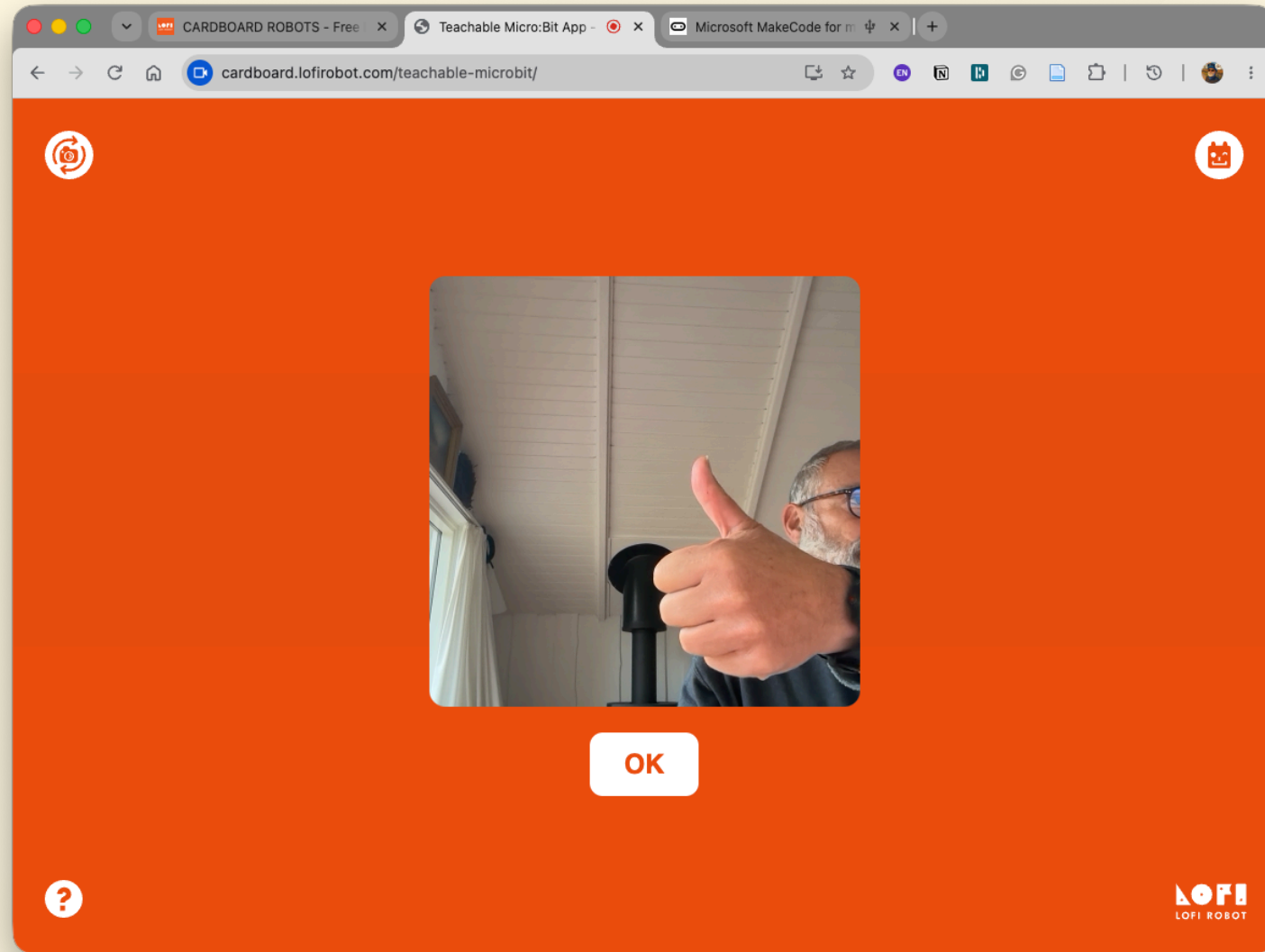
Model Select



Pairing



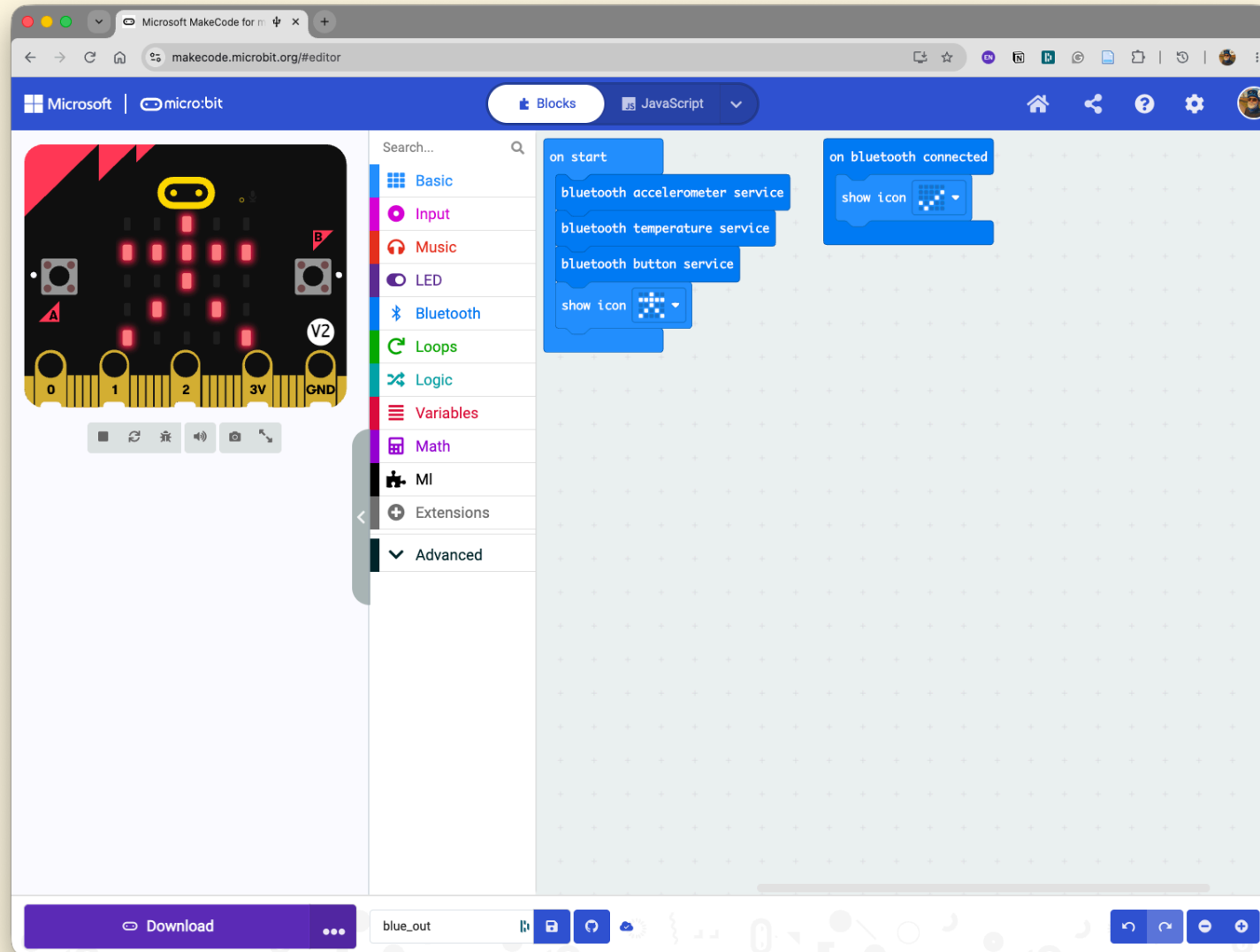
Lofi APP



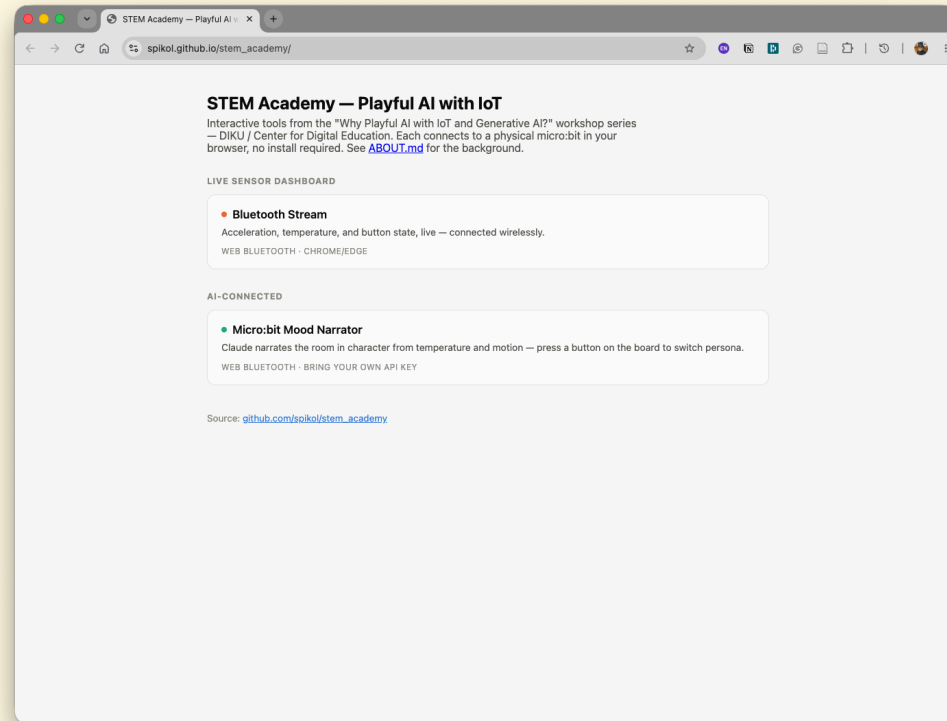
Play Time

micro:bit & LLMs

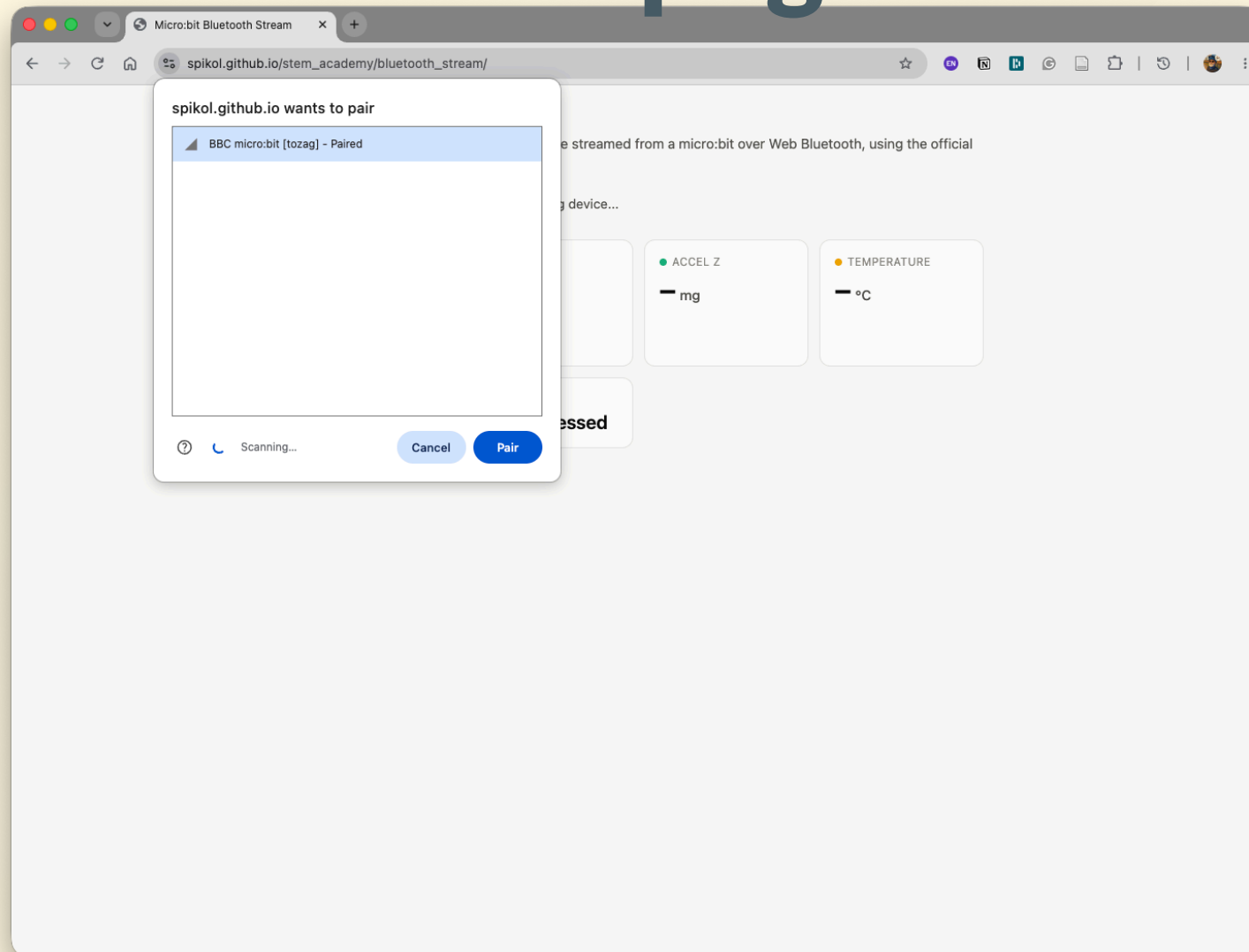
micro:bit code



LLM page https://spikol.github.io/stem_academy/



LLM page



How It Works



API Key - See PADLET

